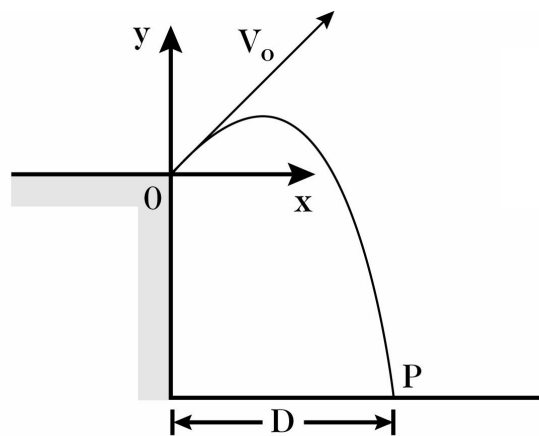


MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) A boy throws a rock with an initial velocity of 2.15 m/s at  $30.0^\circ$  above the horizontal. If air resistance is negligible, how long does it take for the rock to reach the maximum height of its trajectory? 1) \_\_\_\_\_  
 hint: the initial vertical velocity is  $2.15 \sin(30)$ . At the top, the vertical velocity is 0.  
 You can use  $a = (V_f - V_o) / t$   
 A) 0.110 s                      B) 0.215 s                      C) 0.194 s                      D) 0.303 s
- 2) A polar bear starts at the North Pole. It travels 1.0 km south, then 1.0 km east, and then 1.0 km north to return to its starting point. This trip takes 45 min. What was the bear's average velocity? 2) \_\_\_\_\_  
 A) 0.00 km/h                      B) 5.3 km/h                      C) 0.067 km/h                      D) 4.0 km/h
- 3) An object is thrown upwards with a speed of 16 m/s. How long does it take it to reach a height of 7.0 m on the way up? Neglect air resistance. 3) \_\_\_\_\_  
 A) 2.4 s                      B) 3.1 s                      C) 4.2 s                      D) 0.52 s                      E) 1.2 s
- 4) A projectile is fired at time  $t = 0.0$ s, from point 0 at the edge of a cliff, with initial velocity components of  $v_{ox} = 30$  m/s and  $v_{oy} = 300$  m/s. The projectile rises, then falls into the sea at point P. The time of flight of the projectile is 75.0 s. 4) \_\_\_\_\_

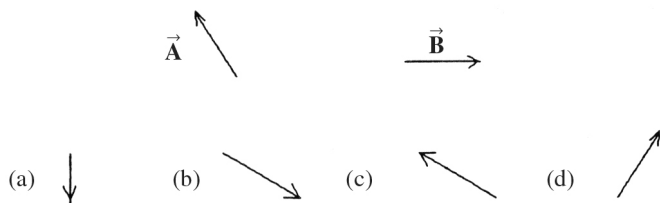
Figure 3.2d



In Figure 3.2d, the horizontal distance D is closest to:

- A) 2,790 m                      B) 3,330 m                      C) 2,520 m                      D) 3,060 m                      E) 2,250 m
- 5) What is the maximum distance we can shoot a dart, from ground level, provided our toy dart gun gives a maximum initial velocity of 2.78 m/s and air resistance is negligible? 5) \_\_\_\_\_  
 A) 1.58 m                      B) 0.394 m                      C) 0.789 m                      D) 1.39 m

- 6) A polar bear starts at the North Pole. It travels 1.0 km south, then 1.0 km east, and then 1.0 km north to return to its starting point. This trip takes 45 min. What was the bear's average speed? 6) \_\_\_\_\_  
 A) 0.00 km/h B) 4.0 km/h C) 5.3 km/h D) 0.067 km/h
- 7) A motorist travels for 3.0 h at 80 km/h and 2.0 h at 100 km/h. What is her average speed for the trip? 7) \_\_\_\_\_  
 A) 85 km/h B) 92 km/h C) 88 km/h D) 90 km/h
- 8) A car accelerates from 5.0 m/s to 21 m/s at a constant rate of 3.0 m/s<sup>2</sup>. How far does it travel while accelerating? 8) \_\_\_\_\_  
 A) 117 m B) 41 m C) 207 m D) 69 m
- 9) The position of an object as a function of time is given by  $x = bt^2 - ct$ , where  $b = 2.0 \text{ m/s}^2$  and  $c = 6.7 \text{ m/s}$ , and  $x$  and  $t$  are in SI units. What is the instantaneous velocity of the object when  $t = 2.2$ ? 9) \_\_\_\_\_  
 A) 2.3 m/s B) 2.7 m/s C) 1.7 m/s D) 2.1 m/s
- 10) Consider two vectors  $\vec{A}$  and  $\vec{B}$  shown in the figure. The difference  $\vec{A} - \vec{B}$  is best illustrated by 10) \_\_\_\_\_



- A) choice (a). B) choice (b). C) choice (c). D) choice (d).

## Answer Key

Testname: TEST1\_REVIEW

- 1) A
- 2) A
- 3) D
- 4) E
- 5) C
- 6) B
- 7) C
- 8) D
- 9) D
- 10) C